Apple A20 Pro brings 2nm process technology to phones

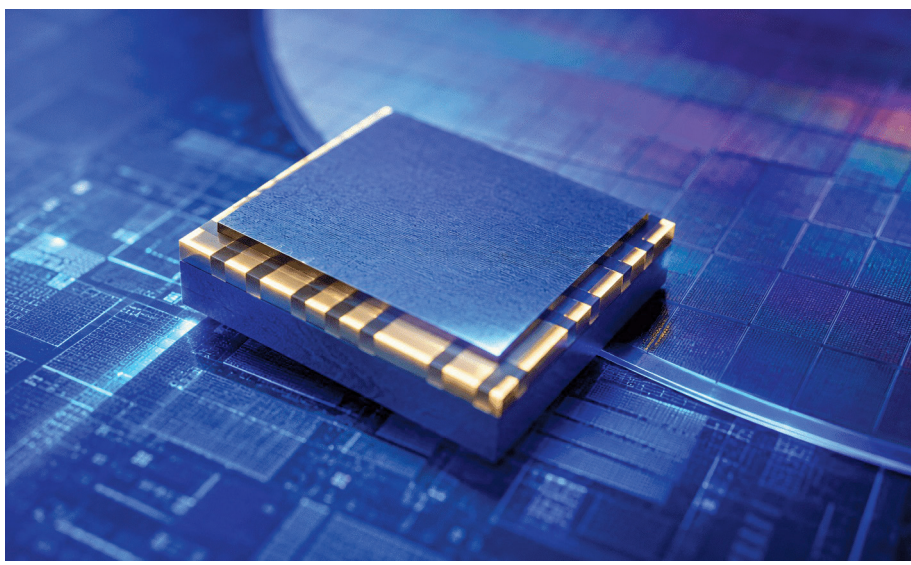


The A20 Pro brings gate-all-around nanosheet transistors, a faster Neural Engine and improved thermal design to Apple's latest devices (Credit: Apple)

Apple's A20 Pro processor brings gate-all-around (GAA) nanosheet transistors to the company's latest smartphones, marking a shift in transistor architecture beyond FinFET technology. Built on a 2-nanometre process generation, the chip targets higher performance, density, and power efficiency. Apple reports up to 20 per cent faster CPU performance and 40 per cent faster GPU performance, alongside a faster Neural Engine and increased memory bandwidth. The processor also uses an advanced package and vapour-chamber cooling system developed for sustained performance in demanding AI and mobile workloads.

Low-resistance contacts improve GaN electronics

A new processing technique developed at Nagoya University in Japan significantly reduces electrical resistance in thin p-type gallium nitride (p-GaN) semiconductor devices. By depositing an ultrathin magnesium layer under 10 nanometres thick onto the p-GaN surface and applying a five-minute soft annealing treatment at 600°C, researchers achieved an ultra-low contact resistivity of (1-3) × 10^{-4} Ω cm^2 while maintaining structural smoothness. Unlike complex bottom-up crystal growth methods, this top-down approach offers a simpler, faster, and cheaper integration path that can be applied post-fabrication. The breakthrough could substantially minimise energy losses in high-efficiency LEDs and power transistors deployed across electric vehicles and data centres.



New technique reduces electrical resistance in thin p-GaN devices significantly (Credit: https://en.nagoya-u.ac.jp)